

The screenshot shows a web browser window with the SIMATS Online Programming Lab interface. The page title is "SIMATS Online Programming Lab" and the user is logged in as "R HARSHA KYLE". The assignment is titled "CO2-AT2-Assignment" and is worth 10 marks. The question is "Q1 Inventory Product Search System". The problem statement is: "A supermarket stores product names in different letter cases. Develop a program that: Converts product names to lowercase using toLowerCase(); Searches a product using contains(); Confirms exact matching using equalsIgnoreCase()". The user's code is shown in a text area, and the output is displayed in a box on the right.

QUESTIONS

- Q1 Inventory Product Search System 10 marks
- Q2 Hospital Appointment Scheduling System 10 marks
- Q3 Bank Token Management System 10 marks

3/3 answered

Q1 Inventory Product Search System 10 marks

A supermarket stores product names in different letter cases. Develop a program that:

- Converts product names to lowercase using toLowerCase()
- Searches a product using contains()
- Confirms exact matching using equalsIgnoreCase()

Your Code

```
1 public class Main{
2     public static void main(String[] args){
3         String p1 ="Milk Packet";
4         String p2="milk packet";
5         String low = p1.toLowerCase();
6         System.out.println("Lowercase Product Name: "+low);
7         String s="packet";
8         int f;
9         if(low.contains(s)){
10             f=1;
11         }else{
12             f=0;
13         }
14     }
15 }
```

Input

stdin input

Output

```
Lowercase Product Name: milk packet
Product contains 'packet': 1
Exact Match Found: 1
```

```
public class Main{
public static void main(String[] args){
String p1 ="Milk Packet";
String p2="milk packet";
String low = p1.toLowerCase();
System.out.println("Lowercase Product Name: "+low);
String s="packet";
int f;
if(low.contains(s)){
f=1;
}else{
f=0;
}
System.out.println("Product contains 'packet': "+f);
int chk;
if(p1.equalsIgnoreCase(p2)){
chk =1;
}else{
chk=0;
}
System.out.println("Exact Match Found: "+chk);
}
}
```

The screenshot shows a web browser window with the SIMATS Online Programming Lab interface. The page title is "SIMATS Online Programming Lab" and the user is logged in as "R HARSHA KYLE". The assignment is titled "CO2-AT2-Assignment" and is worth 10 marks. The problem description is "Hospital Appointment Scheduling System". The requirements are: "A hospital issues appointment tokens to patients and stores patient details separately. Requirements: Add appointment tokens to a Queue. Store token number and patient name in a HashMap. Serve patients in token order and display the served patient details". The input is: "101 Harsha, 102 Mann, 103 Charlie, 2". The output is: "101 Harsha, 102 Mann". The code is written in Java and uses a Scanner, Queue, and HashMap.

QUESTIONS

- Q1 Inventory Product Search System 10 marks
- Q2 Hospital Appointment Scheduling System 10 marks
- Q3 Bank Token Management System 10 marks

3/3 answered

Q2 Hospital Appointment Scheduling System 10 marks

A hospital issues appointment tokens to patients and stores patient details separately.

Requirements:

- Add appointment tokens to a Queue
- Store token number and patient name in a HashMap
- Serve patients in token order and display the served patient details

Your Code

```
1 import java.util.*;
2 public class Main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         Queue<Integer> q = new LinkedList<Integer>();
8         HashMap<Integer,String> mp = new HashMap<Integer,String>();
9         int n = sc.nextInt();
10        for(int i=0;i<n;i++)
11        {
```

Input

```
3
101 Harsha
102 Mann
103 Charlie
2
```

Output

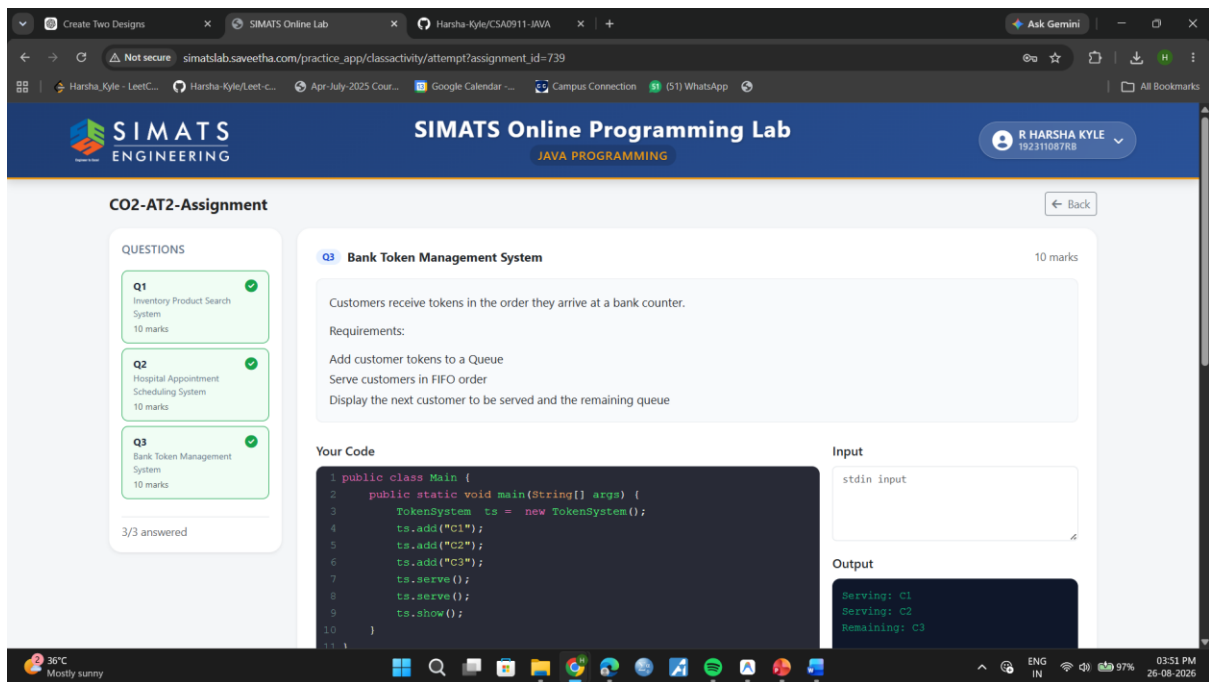
```
101 Harsha
102 Mann
```

```
import java.util.*;
public class Main
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        Queue<Integer> q = new LinkedList<Integer>();
        HashMap<Integer,String> mp = new HashMap<Integer,String>();
        int n = sc.nextInt();
        for(int i=0;i<n;i++)
        {
            int t = sc.nextInt();
            String nm = sc.next();
            q.add(t);
            mp.put(t,nm);
        }
        int k = sc.nextInt();
        for(int i=0;i<k;i++)
        {
            if(q.isEmpty()==false)
            {
                int id = q.poll();
                String name = mp.get(id);
                System.out.println(id+" "+name);
            }
        }
    }
}
```

```

else
{
System.out.println("No more patients");
}
}
}
}
}
}

```



```

public class Main {
public static void main(String[] args) {
TokenSystem ts = new TokenSystem();
ts.add("C1");
ts.add("C2");
ts.add("C3");
ts.serve();
ts.serve();
ts.show();
}
}

```

```

class TokenSystem {
String[] q= new String[10];
int head= 0;
int tail=0;

void add(String t) {

```

```
q[tail] = t;  
tail++;  
}
```

```
void serve() {  
    if (head < tail) {  
        System.out.println("Serving: " + q[head]);  
        head++;  
    } else {  
        System.out.println("No customers");  
    }  
}
```

```
void show() {  
    System.out.print("Remaining: ");  
    int i = head;  
    while (i < tail) {  
        System.out.print(q[i] + " ");  
        i++;  
    }  
    System.out.println();  
}
```